Lab 12 Activity

```r
## the following code should be run first (from the main lab)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.1     v stringr   1.5.2
v ggplot2   4.0.0     v tibble    3.3.0
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.1.0
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```r
library(haven)
library(caret)
```

```
Loading required package: lattice

Attaching package: 'caret'

The following object is masked from 'package:purrr':

    lift
```

```r
library(psych)
```

```
Attaching package: 'psych'
```

```
The following objects are masked from 'package:ggplot2':

    %+%, alpha
```

```r
library(paran)
```

```
Loading required package: MASS

Attaching package: 'MASS'

The following object is masked from 'package:dplyr':

    select
```

```r
library(lavaan)
```

```
This is lavaan 0.6-20
lavaan is FREE software! Please report any bugs.

Attaching package: 'lavaan'

The following object is masked from 'package:psych':

    cor2cov
```

```r
dat <- read_sav("Lab_12_Cox.sav")
```

The dataset that includes the Panas items also includes the Relationship Scales Questionnaire (RSQ), which includes 30 items on 5-point Likert scales (scored 1-5). Although the entire set of 30 items can be thought of as a single scale, the following subscales are also theoretically present:

- RSQ items 11, 18, 21, 23, 25 : attachment avoidance
- RSQ items 10, 12, 13, 15, 20, 24, 29, 30: attachment anxiety

Here, RSQ items 12, 13, 20, 24, 29 need to be reverse coded, which can be done by running the following code:

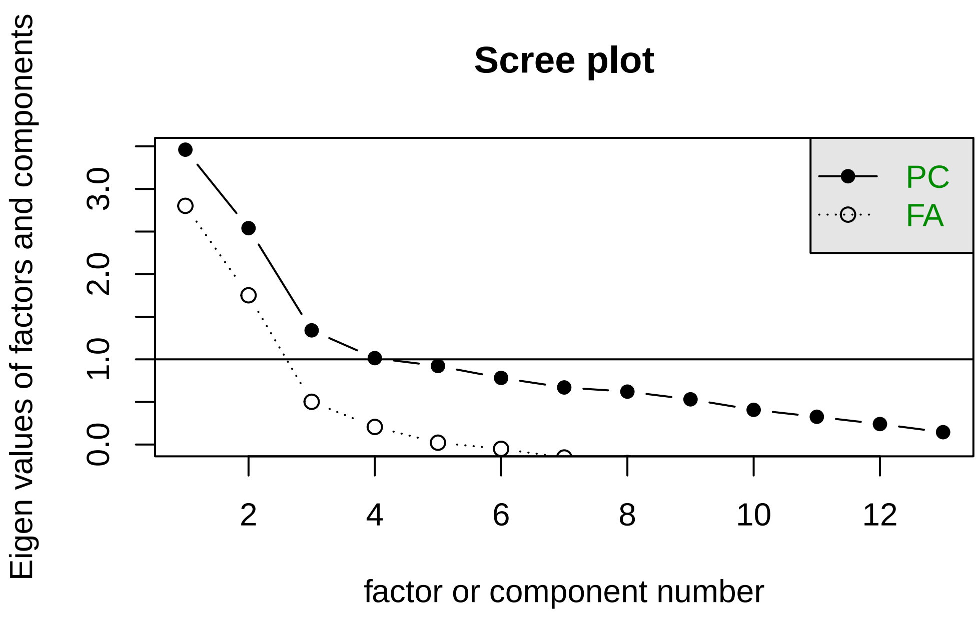
```r
dat <- dat %>% mutate(RSQ12 = 6 - RSQ12, RSQ13 = 6 - RSQ13,
                      RSQ20 = 6 - RSQ20, RSQ24 = 6 - RSQ24,
                      RSQ29 = 6 - RSQ29)
dat_for_fa <- dat %>%
  dplyr::select(RSQ11, RSQ18, RSQ21, RSQ23, RSQ25,
                RSQ10, RSQ12, RSQ13, RSQ15, RSQ20,
                RSQ24, RSQ29, RSQ30)
```

1. First, split the data into a training (exploratory) set and a test (confirmatory) set. Use a different split than used in the lab slides by setting a different seed.

```r
set.seed(255) # answer will vary for this and all results due to different seeds
idx <- createFolds(1:nrow(dat_for_fa), 2)
dat_efa <- dat_for_fa[idx$Fold1, ]
dat_cfa <- dat_for_fa[idx$Fold1, ]
```

2. Create a scree plot and conduct parallel analysis using your exploratory data set. Based on these results, decide on the number of factors to retain.

```r
scree(dat_efa)
```



```r
paran(dat_efa, cfa = FALSE)
```

```
Using eigendecomposition of correlation matrix.
Computing: 10%  20%  30%  40%  50%  60%  70%  80%  90%  100%


Results of Horn's Parallel Analysis for component retention
390 iterations, using the mean estimate

--------------------------------------------------
Component   Adjusted    Unadjusted    Estimated
            Eigenvalue  Eigenvalue    Bias
--------------------------------------------------
1           2.800319    3.461102      0.660783
2           2.062947    2.539711      0.476764
--------------------------------------------------

Adjusted eigenvalues > 1 indicate dimensions to retain.
(2 components retained)
```

```r
paran(dat_efa, cfa = TRUE)
```

```
Using eigendecomposition of correlation matrix.
Computing: 10%  20%  30%  40%  50%  60%  70%  80%  90%  100%


Results of Horn's Parallel Analysis for factor retention
390 iterations, using the mean estimate

--------------------------------------------------
Factor      Adjusted    Unadjusted    Estimated
            Eigenvalue  Eigenvalue    Bias
--------------------------------------------------
No components passed.
--------------------------------------------------
1           2.214956    3.013561      0.798605
2           1.378444    1.987597      0.609153
3           0.274426    0.743089      0.468663
4           0.039089    0.394851      0.355761
```

```
--------------------------------------------------

Adjusted eigenvalues > 0 indicate dimensions to retain.
(4 factors    retained)
```

```
# Answers may vary - I'll proceed with 4
```

3. Conduct an exploratory factor analysis (using the exploratory data set) with the number of factors that you identified in the previous step. Does your EFA support the theoretical dimensions?

```
efa_res <- fa(dat_efa, nfactors = 4, rotate = "oblimin", fm = "pa")
```

```
Loading required namespace: GPArotation
```

```
print(efa_res$loadings, cutoff = .3)
```

```
Loadings:
      PA1    PA2    PA3    PA4
RSQ11  0.878
RSQ18                0.523
RSQ21  0.955
RSQ23  0.739
RSQ25                0.769
RSQ10                0.371
RSQ12         0.502
RSQ13         0.882
RSQ15
RSQ20         0.607
RSQ24         0.602
RSQ29         0.519
RSQ30         0.439  0.301

                 PA1   PA2   PA3   PA4
SS loadings    2.355 2.290 1.198 0.512
Proportion Var 0.181 0.176 0.092 0.039
Cumulative Var 0.181 0.357 0.449 0.489
```

```
# answers may vary
# not the cleanest solution -
#  avoidance relates to PA1 and PA3, anxiety mostly relates to PA2
```

4. What changes (if any) would you make to the EFA? Test out any changes you identify.

```
# what if we only use 2 factors?
efa_res <- fa(dat_efa, nfactors = 2, rotate = "oblimin", fm = "pa")
print(efa_res$loadings, cutoff = .3) # somewhat cleaner solution
```

```
Loadings:
      PA1    PA2
RSQ11  0.786
RSQ18  0.432
RSQ21  0.942
RSQ23  0.732
RSQ25  0.373
RSQ10  0.333  0.329
RSQ12         0.507
RSQ13         0.762
RSQ15
RSQ20         0.653
RSQ24         0.523
RSQ29         0.536
RSQ30         0.567

                 PA1   PA2
SS loadings    2.575 2.435
Proportion Var 0.198 0.187
Cumulative Var 0.198 0.385
```

5. Use CFA with the confirmatory data set to test the theoretical structure of these data (outlined at the top of this assignment), and evaluate fit. Does the model fit the data well?

```
mod <- "Avoid =~ RSQ11 + RSQ18 + RSQ21 + RSQ23 + RSQ25
        Anxiety =~ RSQ10 + RSQ12 + RSQ13 + RSQ15 + RSQ20 + RSQ24 + RSQ29 + RSQ30"
cfa_res <- cfa(model = mod, data = dat_cfa, estimator = "MLM")
standardizedSolution(cfa_res)
```

```
       lhs op     rhs est.std    se      z pvalue ci.lower ci.upper
1    Avoid =~   RSQ11   0.846 0.040 21.356  0.000    0.768    0.924
2    Avoid =~   RSQ18   0.297 0.086  3.455  0.001    0.128    0.465
3    Avoid =~   RSQ21   0.968 0.024 41.056  0.000    0.921    1.014
4    Avoid =~   RSQ23   0.766 0.065 11.876  0.000    0.640    0.893
5    Avoid =~   RSQ25   0.253 0.091  2.795  0.005    0.076    0.430
6  Anxiety =~   RSQ10   0.167 0.115  1.458  0.145   -0.058    0.392
7  Anxiety =~   RSQ12   0.586 0.072  8.130  0.000    0.445    0.728
8  Anxiety =~   RSQ13   0.817 0.053 15.532  0.000    0.714    0.920
9  Anxiety =~   RSQ15   0.102 0.111  0.922  0.357   -0.115    0.320
10 Anxiety =~   RSQ20   0.650 0.087  7.483  0.000    0.480    0.821
11 Anxiety =~   RSQ24   0.586 0.075  7.770  0.000    0.438    0.734
12 Anxiety =~   RSQ29   0.505 0.105  4.793  0.000    0.299    0.712
13 Anxiety =~   RSQ30   0.513 0.134  3.822  0.000    0.250    0.776
14   RSQ11 ~~   RSQ11   0.284 0.067  4.244  0.000    0.153    0.416
15   RSQ18 ~~   RSQ18   0.912 0.051 17.877  0.000    0.812    1.012
16   RSQ21 ~~   RSQ21   0.064 0.046  1.394  0.163   -0.026    0.153
17   RSQ23 ~~   RSQ23   0.413 0.099  4.178  0.000    0.219    0.607
18   RSQ25 ~~   RSQ25   0.936 0.046 20.441  0.000    0.846    1.026
19   RSQ10 ~~   RSQ10   0.972 0.038 25.365  0.000    0.897    1.047
20   RSQ12 ~~   RSQ12   0.656 0.085  7.754  0.000    0.490    0.822
21   RSQ13 ~~   RSQ13   0.333 0.086  3.873  0.000    0.164    0.501
22   RSQ15 ~~   RSQ15   0.990 0.023 43.561  0.000    0.945    1.034
23   RSQ20 ~~   RSQ20   0.577 0.113  5.101  0.000    0.355    0.799
24   RSQ24 ~~   RSQ24   0.657 0.088  7.435  0.000    0.484    0.830
25   RSQ29 ~~   RSQ29   0.745 0.106  6.996  0.000    0.536    0.954
26   RSQ30 ~~   RSQ30   0.737 0.138  5.344  0.000    0.466    1.007
27   Avoid ~~   Avoid   1.000 0.000     NA     NA    1.000    1.000
28 Anxiety ~~ Anxiety   1.000 0.000     NA     NA    1.000    1.000
29   Avoid ~~ Anxiety  -0.218 0.103 -2.104  0.035   -0.420   -0.015
```

```r
fitMeasures(cfa_res,
            fit.measures = c("chisq.scaled", "df.scaled",
                             "pvalue", "rmsea.scaled",
                             "srmr", "cfi.scaled", "tli.scaled"))
```

```
chisq.scaled    df.scaled       pvalue rmsea.scaled         srmr   cfi.scaled
     104.402       64.000        0.000        0.079        0.107        0.877
  tli.scaled
       0.850
```

```
# yet again, not stellar fit
```

6. What changes (if any) would you consider making to the CFA?

```
# Many possible answers, based on the previous results
# Maybe omit some items or allow cross-loadings based on the EFA results
# Or revisit item text to see if some items don't make theoretical sense
```